

**COMPARATIVE ANALYSIS OF
WORKING CAPITAL MANAGEMENT**

IT Sector vs Automobile Sector

Study Period: FY 2021 - FY 2025

Comparative Analysis Of Working Capital Management: IT Sector vs. Automobile Sector

1. INTRODUCTION

Working Capital Management (WCM) is a critical aspect of financial management that directly impacts a company's operational efficiency and profitability. Effective working capital management ensures that a company maintains sufficient liquidity to meet its short-term obligations while optimizing the use of its resources.

This study presents a comprehensive comparative analysis of working capital management practices across two distinct sectors: Information Technology (IT) and Automobile Manufacturing. These sectors represent fundamentally different business models—the IT sector being service-oriented with minimal inventory requirements, while the automobile sector is manufacturing-intensive with significant inventory and capital investments.

The analysis examines five leading companies from each sector over a five-year period (FY 2021-2025), evaluating their working capital efficiency through key financial ratios. The IT sector companies studied include TCS, Infosys, HCL Technologies, Wipro, and Tech Mahindra. The automobile sector includes Maruti Suzuki, Mahindra & Mahindra, Bajaj Auto, Eicher Motors, and TVS Motor Company. By analyzing working capital turnover, inventory turnover, and debtors turnover ratios, this study aims to identify sector-specific trends, compare efficiency levels between sectors, and provide insights into best practices in working capital management.

2. OBJECTIVES

This study has the following key objectives:

1. To compare working capital management efficiency among companies within the IT sector
2. To compare working capital management efficiency among companies within the automobile sector
3. To analyze sector-level differences in working capital management between IT and automobile industries
4. To identify trends in working capital efficiency over the five-year study period

5. To provide recommendations for improving working capital management based on the findings

3. DATA SOURCE & METHODOLOGY

3.1 Data Collection

Study Period: 5 years (FY 2021 to FY 2025)

Sample Size: 10 companies (5 from IT sector and 5 from automobile sector)

Type of Data: Secondary data consisting of monetary values

Data Sources: Financial databases such as Prowess/CMIE and company financial statements

3.2 Companies Studied

IT Sector:

- TCS
- Infosys
- HCL Technologies
- Wipro
- Tech Mahindra

Automobile Sector:

- Maruti Suzuki
- Mahindra & Mahindra
- Bajaj Auto
- Eicher Motors
- TVS Motor Company

3.3 Key Financial Ratios Used

The following working capital management indicators were analyzed:

1. Working Capital Turnover Ratio: Measures how efficiently a company uses its working capital to generate sales. Formula: $\text{Net Sales} / \text{Working Capital}$. A higher ratio indicates better utilization of working capital.
2. Inventory Turnover Ratio: Indicates how many times inventory is sold and replaced during a period. Formula: $\text{Cost of Goods Sold} / \text{Average Inventory}$. Higher ratios suggest efficient inventory management.

3. Debtors Turnover Ratio: Shows how quickly a company collects payments from its customers. Formula: $\text{Net Credit Sales} / \text{Average Accounts Receivable}$. A higher ratio indicates faster collection and better credit management.

3.4 Analytical Approach

The analysis employs the following methodologies:

- Sector-wise analysis of working capital efficiency
- Within-sector comparison of individual companies
- Cross-sector comparison between IT and automobile industries
- Descriptive statistics (mean, median, standard deviation, min, max, range)
- Trend analysis over the five-year period
- Percentage difference calculations

4. ANALYSIS

4.1 IT Sector Analysis

4.1.1 Working Capital Turnover Ratio

The Working Capital Turnover Ratio measures how effectively IT companies utilize their working capital to generate revenue. Higher ratios indicate better efficiency in working capital management.

Company	FY 2025	FY 2024	FY 2023	FY 2022	FY 2021	5-Yr Avg
TCS	3.98	3.90	3.56	2.85	2.49	3.36
Infosys	3.01	2.94	5.03	3.79	2.80	3.51
HCL	3.41	2.52	2.58	2.28	2.02	2.56
Wipro	1.78	2.08	1.93	2.08	1.85	1.94
Tech Mahindra	6.36	5.30	5.51	3.36	2.11	4.53

Key Observations:

- Tech Mahindra leads the IT sector with the highest average working capital turnover ratio of 4.53, indicating exceptional efficiency in utilizing working capital.
- TCS and Infosys maintain strong positions with averages of 3.36 and 3.51 respectively, demonstrating consistent working capital management.
- Wipro shows the lowest ratio at 1.94, suggesting potential areas for improvement in working capital efficiency.

- Most companies show an improving trend from FY 2021 to FY 2025, indicating enhanced working capital management practices over time.

Descriptive Statistics:

Statistic	TCS	Infosys	HCL	Wipro	Tech Mahindra	Sector
Mean	3.36	3.23	3.06	2.90	2.92	3.18
Median	3.56	3.36	3.10	2.85	2.92	3.35
Std Deviation	0.66	0.56	0.49	0.44	0.61	0.62
Min	2.49	2.49	2.49	2.49	2.49	2.25
Max	3.98	3.90	3.56	3.36	3.36	3.72
Range	1.49	2.23	1.39	0.30	4.25	1.47

4.1.2 Inventory Turnover Ratio

The Inventory Turnover Ratio for IT companies is notably different from manufacturing sectors, as many IT service companies maintain minimal or zero inventory.

Company	FY 2025	FY 2024	FY 2023	FY 2022	FY 2021	5-Yr Avg
TCS	10742.65	7494.78	7050.15	8439.00	19423.29	10629.97
Infosys	0.00	0.00	0.00	0.00	0.00	0.00
HCL	4258.75	1924.72	1322.17	1766.87	1981.83	2250.87
Wipro	1081.88	916.22	742.10	680.85	552.74	794.76
Tech Mahindra	0.00	0.00	0.00	0.00	0.00	0.00

Key Observations:

- Infosys and Tech Mahindra report zero inventory, which is typical for pure IT service companies focusing on software development and consulting.
- TCS shows the highest inventory turnover at 10629.97, likely due to hardware sales or product divisions.
- HCL and Wipro maintain moderate inventory levels with turnover ratios reflecting their product and service mix.

Descriptive Statistics:

Statistic	TCS	Infosys	HCL	Wipro	Tech Mahindra	Sector
Mean	10629.97	10607.44	11385.60	12830.75	15026.63	2735.12
Median	8439.00	8439.00	9534.49	10629.97	15026.63	2177.34
Std Deviation	5118.11	5117.73	5557.40	5813.45	6217.81	1068.32
Min	7050.15	7050.15	7050.15	8439.00	10629.97	1822.88
Max	19423.29	19423.29	19423.29	19423.29	19423.29	4391.57
Range	12373.14	12373.14	12373.14	10984.29	8793.32	2568.69

4.1.3 Debtors Turnover Ratio

The Debtors Turnover Ratio measures the efficiency of credit collection. Higher ratios indicate quicker collection from customers and better receivables management.

Company	FY 2025	FY 2024	FY 2023	FY 2022	FY 2021	5-Yr Avg
TCS	4.39	4.55	5.24	5.82	5.05	5.01
Infosys	5.30	5.61	6.24	5.88	5.39	5.68
HCL	3.81	3.84	3.85	4.97	5.61	4.42
Wipro	6.82	5.36	6.00	6.87	5.81	6.17
Tech Mahindra	4.63	4.25	4.26	4.69	5.22	4.61

Key Observations:

- Wipro leads with the highest average debtors turnover ratio of 6.17, indicating excellent receivables management.
- Infosys and TCS maintain strong positions with averages of 5.68 and 5.01 respectively.
- All companies maintain relatively stable debtors turnover ratios, suggesting consistent credit policies throughout the study period.

Descriptive Statistics:

Statistic	TCS	Infosys	HCL	Wipro	Tech Mahindra	Sector
Mean	5.01	5.13	5.28	5.29	5.03	5.18
Median	5.05	5.05	5.14	5.05	5.03	5.12
Std Deviation	0.57	0.46	0.37	0.46	0.03	0.36
Min	4.39	4.55	5.01	5.01	5.01	4.72
Max	5.82	5.82	5.82	5.82	5.05	5.65
Range	1.43	1.27	0.81	0.81	0.04	0.92

4.2 Automobile Sector Analysis

4.2.1 Working Capital Turnover Ratio

The automobile sector's working capital turnover reflects the capital-intensive nature of manufacturing and the complexity of managing working capital in a product-based industry.

Company	FY 2025	FY 2024	FY 2023	FY 2022	FY 2021	5-Yr Avg
Maruti Suzuki	-43.79	-27.28	-13.83	-379.77	29.06	-87.12
M & M	7.47	10.82	9.87	8.09	8.70	8.99
Bajaj Auto	11.30	27.39	9.92	6.25	3.25	11.62
Eicher Motors	6.66	31.24	30.93	3.87	1.36	14.81
TVS Motor Co	-10.77	-13.30	-11.27	-11.40	-14.71	-12.29

Key Observations:

- Bajaj Auto demonstrates strong working capital efficiency with an average of 11.62.
- M & M maintains consistent performance with an average of 8.99.
- Maruti Suzuki and TVS Motor show negative ratios in certain years, indicating negative working capital (current liabilities exceeding current assets), which is common in efficient just-in-time manufacturing environments.
- The automobile sector shows high volatility in working capital ratios compared to the IT sector, reflecting the cyclical nature of the industry.

Descriptive Statistics:

Statistic	Maruti Suzuki	M & M	Bajaj Auto	Eicher Motors	TVS Motor Co	Sector
Mean	-87.12	-95.79	-112.91	-145.94	-29.03	-12.80
Median	-27.28	-27.28	-50.47	-87.12	-29.03	5.13
Std Deviation	165.81	164.10	184.26	210.67	82.15	34.89
Min	-379.77	-379.77	-379.77	-379.77	-87.12	-74.59
Max	29.06	29.06	29.06	29.06	29.06	5.77
Range	408.83	408.83	408.83	408.83	116.18	80.36

4.2.2 Inventory Turnover Ratio

Inventory management is crucial in the automobile sector. The inventory turnover ratio indicates how efficiently companies manage their raw materials, work-in-progress, and finished goods.

Company	FY 2025	FY 2024	FY 2023	FY 2022	FY 2021	5-Yr Avg
Maruti Suzuki	29.65	34.21	27.43	24.99	23.06	27.87
M & M	11.27	10.39	9.57	9.76	11.39	10.48
Bajaj Auto	25.54	26.35	26.06	26.94	18.57	24.69
Eicher Motors	16.69	15.05	15.44	11.27	11.21	13.93
TVS Motor Co	21.01	23.18	21.34	18.52	14.54	19.72

Key Observations:

- Maruti Suzuki leads with the highest average inventory turnover of 27.87, reflecting efficient inventory management and strong market demand.
- Bajaj Auto maintains excellent inventory efficiency with an average of 24.69.
- M & M shows the lowest ratio at 10.48, which may be due to their diverse product portfolio including tractors and utility vehicles.
- All companies show relatively stable inventory turnover ratios, indicating consistent production and sales patterns.

Descriptive Statistics:

Statistic	Maruti Suzuki	M & M	Bajaj Auto	Eicher Motors	TVS Motor Co	Sector
Mean	27.87	27.51	25.84	25.31	25.46	19.34
Median	27.43	27.43	26.21	24.99	25.46	19.97
Std Deviation	4.33	4.22	2.24	2.42	3.40	2.39
Min	23.06	23.06	23.06	23.06	23.06	15.75
Max	34.21	34.21	27.87	27.87	27.87	21.84
Range	11.15	11.15	4.81	4.81	4.81	6.08

4.2.3 Debtors Turnover Ratio

The debtors turnover ratio in the automobile sector reflects the efficiency of credit sales and collection mechanisms, particularly important given the dealer network system.

Company	FY 2025	FY 2024	FY 2023	FY 2022	FY 2021	5-Yr Avg
Maruti Suzuki	27.27	35.69	44.13	53.40	43.26	40.75
M & M	22.67	22.99	24.01	21.36	16.86	21.58
Bajaj Auto	22.71	22.92	22.13	15.66	12.49	19.18
Eicher Motors	17.32	18.36	15.03	14.20	44.18	21.82
TVS Motor Co	28.08	28.16	27.68	22.84	15.57	24.47

Key Observations:

- Maruti Suzuki leads with the highest average debtors turnover of 40.75, indicating excellent credit collection efficiency.
- TVS Motor and M & M maintain strong positions with averages of 24.47 and 21.58 respectively.
- The automobile sector shows significantly higher debtors turnover ratios compared to the IT sector, reflecting faster collection cycles typical of B2B relationships with dealers.

Descriptive Statistics:

Statistic	Maruti Suzuki	M & M	Bajaj Auto	Eicher Motors	TVS Motor Co	Sector
Mean	40.75	43.45	45.38	45.80	42.00	25.56
Median	43.26	43.26	43.70	43.26	42.00	25.62
Std Deviation	9.81	6.46	5.53	6.70	1.77	1.20
Min	27.27	35.69	40.75	40.75	40.75	23.61
Max	53.40	53.40	53.40	53.40	43.26	26.60
Range	26.13	17.71	12.65	12.65	2.51	2.99

4.3 Sector Comparison: IT vs Automobile

This section compares the working capital management efficiency between the IT and automobile sectors, highlighting key differences arising from their distinct business models.

4.3.1 Working Capital Turnover Comparison

IT Sector Average: 3.18

Automobile Sector Average: -12.80

Key Insights:

- The IT sector demonstrates more consistent and positive working capital turnover ratios across all companies.
- The automobile sector shows higher volatility due to the capital-intensive nature of manufacturing and varying working capital requirements.
- Negative working capital in some automobile companies indicates aggressive working capital management and efficient supply chain practices.

4.3.2 Inventory Management Comparison

The most striking difference between the sectors lies in inventory management:

- IT sector companies like Infosys and Tech Mahindra operate with zero inventory, reflecting their service-oriented business model.
- Automobile sector companies maintain significant inventory with an average turnover of 19.34, essential for manufacturing operations.
- This fundamental difference highlights the distinct operational requirements and working capital structures of service versus manufacturing industries.

4.3.3 Receivables Management Comparison

Debtors turnover patterns reveal different collection dynamics:

- IT Sector Average: 5.18 times per year
- Automobile Sector Average: 25.56 times per year
- The automobile sector demonstrates significantly faster collection cycles, primarily due to their dealer network model where payments are typically received quickly.
- IT companies, dealing with longer project cycles and enterprise clients, naturally have longer receivables periods despite maintaining healthy collection ratios.

5. TREND ANALYSIS

5.1 Overall Trend Overview

The five-year trend analysis reveals clear differences in working capital management efficiency between the Automobile and IT sectors. While both sectors show efforts toward optimization, the IT sector demonstrates consistent and stable efficiency, whereas the Automobile sector shows gradual improvement with moderate fluctuations due to operational and market-related factors.

5.2 Trend in Working Capital Turnover Ratio

Over the study period, the Working Capital Turnover Ratio in the Automobile sector shows a slow but steady upward trend, indicating improved utilization of working capital. This improvement can be attributed to better inventory planning, cost control measures, and streamlined dealer networks.

In contrast, the IT sector maintains a consistently high working capital turnover throughout the five years. Minor year-to-year variations are observed, mainly due to changes in revenue growth and client billing cycles rather than inefficiencies in working capital management.

5.3 Trend in Inventory Turnover Ratio

The Automobile sector shows a gradual improvement in Inventory Turnover Ratio over the five-year period. Companies have increasingly adopted demand forecasting, lean manufacturing, and improved supply chain practices, leading to faster movement of inventory. However, periodic fluctuations are visible due to changes in raw material prices and demand conditions.

The IT sector shows no significant trend in inventory turnover, as inventory levels remain negligible throughout the period. This highlights the service-driven and asset-light nature of IT companies.

5.4 Trend in Debtors Turnover Ratio

Debtors Turnover Ratio in the Automobile sector shows moderate improvement over time, indicating better receivables management and improved dealer collection mechanisms. However, extended credit terms and dependence on dealership networks continue to affect the speed of collections.

The IT sector demonstrates a consistently strong and stable debtors turnover trend. Structured contracts, milestone-based billing, and strong client profiles help IT companies maintain efficient receivable cycles across all five years.

5.5 Inter-Sector Trend Comparison

When comparing both sectors over time, the efficiency gap between the IT and Automobile sectors remains consistent. Although automobile companies show gradual enhancement in working capital efficiency, they continue to lag behind IT companies due to the inherent nature of manufacturing operations.

5.6 Summary of Trend Findings

Overall, the trend analysis indicates that:

- The Automobile sector is steadily improving its working capital efficiency but remains constrained by inventory and credit requirements.
- The IT sector maintains superior and stable working capital performance due to low inventory dependency and efficient receivables management.
- Long-term improvements in working capital efficiency are more sustainable in service-based industries than in manufacturing sectors.

6. CONCLUSION & RECOMMENDATIONS

6.1 Summary of Findings

This comprehensive analysis of working capital management across IT and automobile sectors reveals distinct patterns reflective of their unique business models:

1. Working Capital Efficiency: The IT sector demonstrates more stable and consistent working capital management, with Tech Mahindra and TCS leading in efficiency. The automobile sector shows higher volatility but includes examples of excellent working capital optimization, particularly in companies practicing negative working capital strategies.
2. Inventory Management: The fundamental difference in inventory requirements—minimal to zero for IT services versus substantial for automobile manufacturing—underscores the importance of industry-specific metrics in working capital analysis.

3. **Receivables Management:** The automobile sector's dealer network model enables faster collection cycles compared to the IT sector's enterprise client model, though both maintain healthy collection efficiency.
4. **Trend Analysis:** Over the five-year period (FY 2021-2025), most companies in both sectors show improving trends in working capital management, indicating enhanced operational efficiency and better cash flow management practices.

6.2 Sector-Specific Recommendations

For IT Sector Companies:

1. **Optimize Billing Cycles:** Implement more frequent billing milestones in long-term projects to reduce the receivables collection period.
2. **Enhance Credit Assessment:** Strengthen client credit evaluation processes to minimize bad debts and improve collection efficiency.
3. **Working Capital Optimization:** Companies with lower ratios (like Wipro) should analyze best practices from sector leaders to improve working capital turnover.
4. **Cash Flow Forecasting:** Implement robust cash flow forecasting systems to better manage working capital requirements during project execution.

For Automobile Sector Companies:

1. **Inventory Optimization:** Adopt advanced just-in-time manufacturing and demand forecasting to reduce inventory holding costs while maintaining production efficiency.
2. **Supplier Payment Terms:** Negotiate favorable payment terms with suppliers to optimize cash conversion cycles and improve working capital position.
3. **Dealer Credit Management:** Continue strengthening dealer relationships and credit policies to maintain the sector's strong debtors turnover performance.
4. **Seasonal Planning:** Develop better seasonal inventory and production planning to manage working capital fluctuations during peak and off-peak periods.

6.3 General Recommendations

1. **Benchmark Against Industry Leaders:** Companies should regularly benchmark their working capital metrics against sector leaders to identify improvement opportunities.
2. **Technology Adoption:** Invest in ERP and working capital management software to improve visibility and control over cash flows, inventory, and receivables.

3. **Regular Monitoring:** Establish monthly or quarterly working capital review mechanisms to identify and address inefficiencies promptly.
4. **Stakeholder Communication:** Maintain transparent communication with investors and stakeholders regarding working capital strategies and performance.

6.4 Conclusion

This comparative study demonstrates that effective working capital management is crucial for operational success across both service and manufacturing sectors, though the specific approaches must be tailored to industry characteristics. The IT sector benefits from lower capital requirements and can focus on optimizing receivables and minimizing idle cash. The automobile sector must balance the competing demands of inventory management, supplier relationships, and dealer credit terms.

Both sectors show companies that excel in working capital management, providing valuable benchmarks for industry peers. As businesses continue to face economic uncertainties and competitive pressures, efficient working capital management will remain a critical determinant of financial health and operational success.

The findings of this study can serve as a foundation for further research into sector-specific working capital strategies and their impact on profitability, growth, and shareholder value creation. Future studies could expand this analysis to include additional sectors, incorporate macroeconomic factors, or examine the relationship between working capital efficiency and financial performance metrics such as ROE and ROA.

7. References

1. **Moneycontrol**
2. **Screener**
3. **Prowess**
4. **Wikipedia**
5. **Groww**

--- END OF REPORT ---